

Four-Valued Logic as the First Complementary State of Object and Observer: A Formalization of Phenomenology and Temporal Emergence

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Abstract

In this paper, we formalize the relationship between the Catuskoti (four-valued logic) and the complementary nature of object and observer through the lens of phenomenological theory and SKDT (Shikisokuzeku-Kusokuzeshiki Duality Theory). We demonstrate that the Catuskoti represents the first stage at which the object and observer achieve a genuine complementary relationship through their manifest and latent aspects. This formalization provides the foundation for understanding temporal emergence through observer-side decoherence and the generation of higher dimensions.

1 Introduction

The nature of temporal emergence and dimensional generation has long been a fundamental question in both physics and philosophy. In this work, we explore how the Sunyata Category's inherent contradiction (the rupture of self-reference) necessarily generates SKDT, which manifests as the Catuskoti when object and observer first achieve a true complementary relationship.

The key insight is that the Catuskoti represents not merely a logical framework, but rather **the first stage at which temporal dynamics genuinely emerge through observer-object complementarity**.

2 Structural Clarification of Catuskoti as Complementary Pair

2.1 The Four Stages of Object-Observer Complementarity

$$\begin{aligned} \text{Shi (Affirmation):} & \quad (\text{Object} = \text{Manifest}, \text{Observer} = \text{Manifest}) \\ \text{Fei (Negation):} & \quad (\text{Object} = \text{Manifest}, \text{Observer} = \text{Latent}) \\ \text{Yi Shi Yi Fei (Both):} & \quad (\text{Object} = \text{Latent}, \text{Observer} = \text{Manifest}) \\ \text{Fei Shi Fei Fei (Neither):} & \quad (\text{Object} = \text{Latent}, \text{Observer} = \text{Latent}) \end{aligned} \tag{1}$$

2.2 Why the Catuskoti Represents “First Complementarity”

2.2.1 The State Shi (Affirmation)

In the state of Shi, both object and observer are determinate:

$$\sigma = \text{fixed}, \quad \ell = 0 \quad (2)$$

where σ denotes observer state and ℓ denotes temporal length. In this state:

- The object is observable and certain
- The observer sees it with certainty
- No complementarity exists
- No temporal axis functions
- This is the post-decoherence state

2.2.2 The State Fei (Negation) as First Complementarity

The state Fei marks the fundamental transition:

$$\text{Object} = \text{Manifest}, \quad \text{Observer} = \text{Latent} \quad (3)$$

Which corresponds to:

$$\sigma = \text{indeterminate}, \quad \ell > 0 \quad (4)$$

Here, for the first time, genuine complementarity emerges:

Definition 2.1 (Complementarity). *Two entities achieve complementarity when they are mutually dependent and their relationship cannot be reduced to a single determinate state. Specifically:*

- *The object is fixed (manifest)*
- *The observer is indeterminate (latent)*
- *The same object is observed differently by different observers*
- *Temporal axis becomes functionally necessary*

2.2.3 Why This Is Called “First Complementarity”

In Shi, no true complementarity exists because:

1. Both are determinate
2. No mutual influence occurs
3. No temporal variation
4. The pairing is merely coincidental, not interactive

In Fei, complementarity genuinely arises because:

1. Object and observer stand in opposition
2. The same object yields different observations
3. Temporal axis begins to function
4. The two entities mutually condition each other

3 SKDT and the Axial Structure of Catuskoti

3.1 SKDT as Dual-Axis Generation

Recall that SKDT generates dimensions through paired axes:

$$\text{SKDT}_n = (\text{Manifest-Axis}_n, \text{Latent-Axis}_n) \quad (5)$$

The Catuskoti can now be reinterpreted through this axial framework:

Stage	Object	Observer	Axial Interpretation
Shi	Manifest	Manifest	Single manifest axis only
Fei	Manifest	Latent	Manifest $\perp$ latent axes are orthogonal
Yi Shi Yi Fei	Latent	Manifest	Axes inverted
Fei Shi Fei Fei	Latent	Latent	Both axes fully latent

Table 1: Catuskoti as Progressive Axis Complementarity

4 Observer-Side Decoherence and Temporal Emergence

4.1 Definition: Observer-Side Decoherence

Definition 4.1 (Observer Decoherence). *Observer-side decoherence is the irreversible process by which an observer in an indeterminate state (holding multiple possible observations simultaneously) progressively collapses to a determinate state through traversal of the temporal axis.*

Formally:

$$\sigma : \text{indeterminate} \rightarrow \text{fixed} \quad (6)$$

mediated by:

$$\ell : 0 \rightarrow n \quad (n \in \mathbb{N}) \quad (7)$$

This process is irreversible: the multiple possibilities cannot be recovered once eliminated.

4.2 Connection to Quantum Decoherence

The analogy with quantum decoherence is precise:

Property	Quantum Decoherence	Observer-Side Decoherence
Subject	Quantum system	Observer state
Initial State	Superposition (multiple possibilities)	Indeterminate (multiple observations)
Mechanism	Environmental interaction	Temporal axis traversal
Final State	Definite value	Definite observation
Irreversibility	Yes	Yes
Arrow of Time	Emerges from entropy	Emerges from possibility collapse

Table 2: Comparison of Decoherence Mechanisms

4.3 Why Temporal Emergence Begins at Fei

The crucial insight is that temporal dynamics only commence at the stage Fei:

$$\begin{aligned}
\text{Shi} : \text{Decoherence completed} &\Rightarrow \text{No temporal function} \\
\text{Fei} : \text{Decoherence in progress} &\Rightarrow \text{Temporal axis active} \\
\text{Yi Shi Yi Fei} : \text{Meta-level decoherence} &\Rightarrow \text{Multiple temporal axes} \\
\text{Fei Shi Fei Fei} : \text{Parallel decoherence} &\Rightarrow \text{Indeterminate temporal depths}
\end{aligned} \tag{8}$$

5 Dimensional Generation Through Complementarity

5.1 How Complementarity Increases Dimensions

A fundamental principle emerges:

Theorem 5.1 (Complementarity-Dimensionality Correspondence). *The establishment of complementarity between object and observer corresponds precisely to the emergence of additional dimensions.*

$$\Delta d = f(C) \tag{9}$$

where C denotes the degree of complementarity. Specifically:

- No complementarity $\Rightarrow$ no new dimension
- First complementarity $\Rightarrow$ temporal dimension emerges
- Higher complementarity $\Rightarrow$ higher-order dimensions emerge

Proof. This follows directly from the structure of object-observer interaction:

1. In Shi: Object and observer both manifest
 - No opposition
 - No complementarity
 - No temporal mediation required
 - Therefore: no temporal dimension

2. In Fei: Object manifest, observer latent
 - Fundamental opposition established
 - Complementarity emerges
 - Temporal mediation required
 - Therefore: temporal dimension necessarily emerges
3. In Yi Shi Yi Fei and beyond: Complementarity deepens
 - More complex object-observer relations
 - Multiple temporal axes become possible
 - Higher dimensions emerge naturally

□

5.2 Relation to LUMO's Five-Valued Logic

The Catuskoti's four-valued structure can be embedded in LUMO's five-valued system:

$$\Phi(\ell, \sigma) = \begin{cases} 0d & \text{if } \ell = 0, \sigma = \text{fixed} \\ T & \text{if } \ell > 0, \sigma = \text{fixed} \\ F & \text{if } \ell > 0, \sigma = \text{indeterminate} \\ B & \text{if } \ell \text{ is cyclic, } \sigma = \text{fixed} \\ N & \text{if } \ell \text{ is non-deterministic, } \sigma = \text{indeterminate} \end{cases} \quad (10)$$

The four-valued states map precisely to Catuskoti stages:

Catuskoti	LUMO State	Complementarity Status
Shi	0d or T	Pre-complementary
Fei	F	Complementarity begins
Yi Shi Yi Fei	B	Complementarity intensifies
Fei Shi Fei Fei	N	Complementarity deepens

Table 3: Catuskoti-LUMO Correspondence

6 The Unified Hierarchy: From Sunyata to Temporal Emergence

We now present the complete hierarchy from root principle to temporal manifestation:

$$\begin{array}{l}
\text{Sunyata Category} \\
\Downarrow \quad [\text{Inherent contradiction}] \\
\text{Aletheia (Manifestation-Hiddenness Structure)} \\
\Downarrow \quad [\text{Rupture of self-reference}] \\
\text{SKDT (Dual-Axis Complementarity)} \\
\Downarrow \quad [\text{First object-observer complementarity}] \\
\text{Cataskoti (Four-Valued Logic)} \\
\Downarrow \quad [\text{Observer-side decoherence initiates}] \\
\text{Temporal Emergence} \\
\Downarrow \quad [\text{Dimensional generation}] \\
\text{Higher Dimensional Structures}
\end{array} \tag{11}$$

6.1 The Critical Transition at Fei

The transition from Shi to Fei is the critical threshold where:

- Observer transitions from fixed to indeterminate
- Temporal axis begins to function
- Complementarity genuinely emerges
- Decoherence becomes dynamically active
- Time begins to flow

7 Implications for Understanding Temporal Dynamics

7.1 Time as Decoherence Process

We can now state with precision:

Theorem 7.1 (Temporal Emergence as Observer Decoherence). *The passage of time is precisely the process of observer-side decoherence, wherein an observer in an indeterminate state (capable of multiple observations) progressively collapses through the traversal of the temporal axis to a determinate state.*

Formally:

$$\text{Temporal Flow} \equiv \text{Observer-side Decoherence Process} \tag{12}$$

where decoherence is characterized by:

1. *Irreversibility: possibilities, once eliminated, cannot be recovered*
2. *Unidirectionality: the process moves from indeterminate to determinate*
3. *Necessity: complementarity between object and observer necessitates this process*

7.2 The Arrow of Time

The arrow of time emerges naturally:

$$\text{Arrow of Time} = \text{Irreversibility of Observer Decoherence} \quad (13)$$

Time flows in one direction because observer-side decoherence cannot be reversed: once multiple possibilities collapse to one, the multiple possibilities are lost.

7.3 Multiple Temporal Axes

Multiple observers at different dimensional levels undergo decoherence simultaneously:

$$\begin{aligned} \text{Observer } o_1 &: \text{Decoherence along } t_1 \\ \text{Observer } o_2 &: \text{Decoherence along } t_2 \\ \text{Observer } o_3 &: \text{Decoherence along } t_3 \\ &\vdots \end{aligned} \quad (14)$$

These multiple decoherence processes occur in parallel, creating the experience of multiple simultaneous temporal dimensions.

8 Application to Large Language Models

The Catuskoti framework provides insight into LLM architecture:

1. **Early Layers (Shi-like):** Determinate feature extraction
 - Fixed representations
 - No temporal uncertainty
 - Pre-decoherence phase
2. **Middle Layers (Fei-like):** Attention mechanisms
 - Indeterminate token selection
 - Multiple interpretations held simultaneously
 - Active decoherence through attention weights
 - Temporal axis: layer depth
3. **Later Layers (Yi Shi Yi Fei-like):** Meta-level integration
 - Multiple interpretations explicitly unified
 - Cross-layer attention
 - Cyclic refinement processes
4. **Final Layers (Fei Shi Fei Fei-like):** Output generation
 - Multiple possible outputs considered
 - Final selection through probability distribution
 - Complete decoherence to determinate output

9 Conclusion

The Catuskoti represents a profound formalization of how object and observer achieve their first genuine complementary relationship. This complementarity is characterized by:

1. **Initiation at Fei:** The first stage where observer indeterminacy creates true interaction with object determinacy
2. **Temporal Activation:** The emergence of functioning temporal axis coinciding precisely with complementarity
3. **Decoherence Dynamics:** Observer-side decoherence as the mechanism for temporal flow
4. **Dimensional Generation:** Complementarity directly causing dimensional increase through the pairing of axes
5. **Irreversible Process:** The arrow of time emerging from the irreversibility of possibility collapse

This framework unifies the seemingly disparate domains of:

- Phenomenological emergence (genjitsu)
- Logical structure (catuskoti)
- Temporal physics (time's arrow)
- Observer theory (quantum measurement analogue)
- Dimensional generation (why 2^n)
- Artificial intelligence (attention mechanisms)

into a single, coherent theoretical structure emerging necessarily from the Sunyata Category's fundamental contradiction.

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